

RENANCE

OSUN STATE UNIVERSITY, OSOGBO

MCB 307 — Past Questions & Materials

by Renance

Course: MCB 307

Institution: Osun State University, Osogbo

Papers in this pack: 1

Levels: 300

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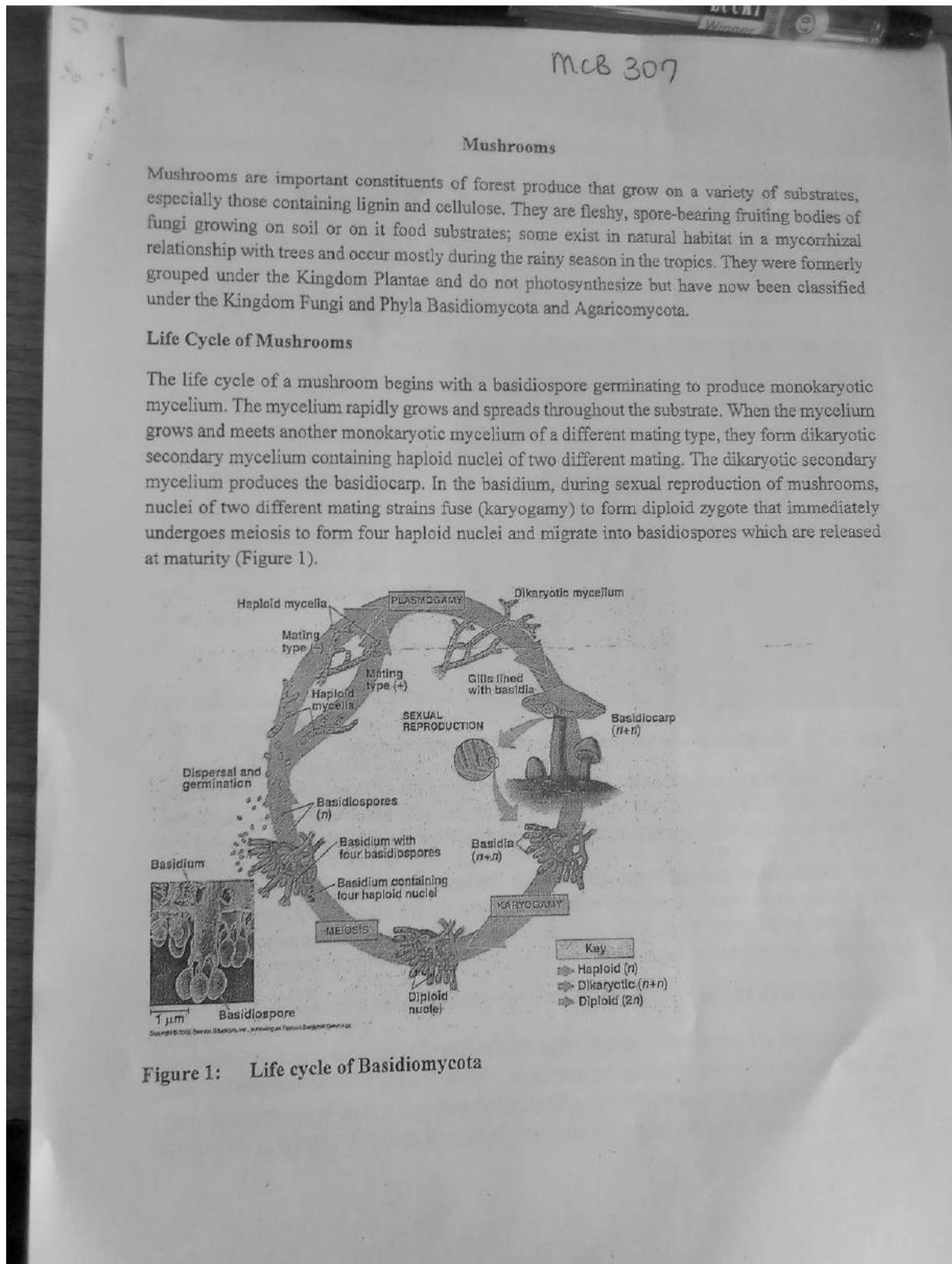
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Paper 1 of 1 — MCB 307 (mushrooms)

Level 300 • First Semester • Handout

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Morphology of Mushrooms

A typical mushroom has the following structures (Figure 2).

- a. **Cap:** This is the top part of a mushroom. It is also known as pileus. On the under surface of the cap, many membranous structures known as gills are present.



Figure 2: Morphology of a tropical mushroom (*Termitomyces* sp.)

- b. **Gills:** These are also called lamellae. They are a series of radially arranged flat surfaces located on the underside of the cap. Each gill has a fertile layer known as hymenium (spore bearing). The hymenium has basidia bearing basidiospores and sterile structures known as paraphyses.
- c. **The stalk:** This is also called the stipe or stem. It may support the cap at the centre, in the middle; it may be off-center (eccentric) or lateral. In some mushrooms, a stalk may be absent. Annulus, also called collar which is the remains of the veil that covers mushroom when young may be present on the stalk of some species.

Distribution of mushrooms

Global assessment of fungi diversity revealed that distribution of mushroom is mainly influenced by climates and certain types of habitat. In contrast to what is obtainable in the temperate zones, mushrooms in the tropics are more diverse and are distributed over a smaller geographical area. In Nigeria, mushrooms occur throughout the year with the highest occurrence mostly in the months of April to June.

Mushrooms are abundant in Southwestern region of Nigeria but almost absent in the Northern parts of the country. Low incidence of mushrooms in the Northern zone was attributed to lengthening desert encroachment, receding vegetation, low annual rainfall and high atmospheric temperatures, non-availability or insufficient suitable nutrient source.

Substrates of mushrooms

Soil debris and dead woods are the most conducive environments for mushroom probably due to high content of degraded nutrients and capacity to retain. Mushrooms have been reportedly found growing on substrates like dead wood and base of dead woods, fallen trees, soil debris, forest and bush areas, grassy places, swamps, palm trees and palm waste, termite nest, cocoa plantations, cassava peels, Fadama areas, and burnt bushes in the wild.

Categories of Mushrooms

a. Mycorrhizal mushroom

They form partnership with trees and some plants. But rather than harming the trees, their presence significantly increases the roots effectiveness. Fungi send their hyphae in and about the little rootlets of the tree until its difficult to tell them apart. The tree supplies the mycelium with moisture and carbohydrates, and the mycelium returns the favour with minerals and other nutrients from the surrounding soil.

b. Parasitic mushroom

This group does serious damage; rather than obtaining their food from the dead plants, they prefer a living host, often attacking and killing. It then live on as saprophytic mushroom.

c. Saprophytic mushroom

They grow on dead organic matter such as fallen trees, cow patties, dead leaves etc. These fungi have enzymes that digest the cellulose and lignin found in the organic matter, with the lignin being an important source of carbon for many organisms. Without their digestive activities, organic material would continue to accumulate until the forest become a huge dump of dead leaves and trees.

Economic Importance of Mushrooms

Mushrooms can be categorized into four major groups based on their economic importance. They are as follows:

a. Edible mushrooms

Edible mushrooms are regarded as meat of the vegetable world. About 200 mushroom species are known to be edible and 25 species widely accepted either as human foods or medicines. In Nigeria,

acceptability of wild edible mushrooms varies from region to region depending on their availability.

i. Identification of edible mushrooms

Several factors such as fruitbody texture, growth substrate, cap texture, colour, odour, strange appearance, and size, are used by laymen to identify and differentiate mushrooms. Most people desist from strange coloured and small sized mushrooms and classify them as inedible or poisonous. In addition, it is believed that when live chicken feed on mushroom without any adverse effect on it, the mushroom is likely to be edible.

Some laymen claimed their ability to differentiate between edible and poisonous mushrooms emanates from their fore-fathers who passed it on to them through verbal communication. Scientific traditional approach of mushroom identification involves morphology description and comparison with the herbarium specimens and photographs or pictorial atlas, and microscopic examination of mushroom mycelia. Generally, the use of the stated parameters cannot be completely relied upon because different mutants and variants of mushrooms having the same morphological structure and appearance may also exist on the same habitat.

b. Medicinal mushrooms

Mushrooms are gaining much attention in pharmaceutical industries, medicines and agro allied companies. Scientific researches on both wild and cultivated mushrooms have shown that they are good therapeutic agents containing several bioactive substances implicated in promoting sound health. Studies on mushrooms and its extracts have shown that mushrooms are possible treatments for diseases. Mushrooms contain bioactive compounds, for example, phenolic compounds, sterols, terpenes, ceramides, glycoproteins, proteoglycans, polysaccharides etc. that are responsible for a great range of healthy properties. Some of these bioactive compounds modulate immune system responses and inhibit tumor growth. Similarly, some mushrooms have exhibited anticholesterol, antitumor, antimicrobial, antiviral, antineoplastic, antimutagenic, antioxidant, antilipidemic, antidiabetic antihyperglycaemic, antihypotensive, antiparasitic, anti-inflammatory hepatoprotective, hypocholesterolemic, immunodulatory and anti-ageing properties.

c. Toxic mushrooms

Some mushroom species produce secondary metabolites that are toxic and result in harmful effect after ingestion. There are only a small number of deadly species; however, several others can cause severe and unpleasant symptoms. Only about thirty-two mushrooms have been associated with fatalities and some identified as containing significant toxins. They classified poisonous mushrooms based on their toxin content, viz: cyclopeptide e.g. amanitoxins and phallotoxins, gyromitrin, orellanine, muscarine, ibotenic acid, coprine, psilocybin and gastrointestinal irritant. These toxins are resistant to destruction by heat.

d. Psychoactive mushrooms

Psychoactive mushrooms have psychedelic properties, i.e. they can cause hallucination and have long played a role in various traditional native medicines worldwide. They have been used as sacraments in rituals and sacrifices aimed at healing mental and physical disabilities, and to facilitate visionary states e.g. *Amanita muscaria*.

e. Other uses of mushrooms

i. Production of biodegradable packaging materials

Mushrooms are currently being employed to make renewable, biodegradable packaging, a direct replacement for the petroleum-based styrofoam. Packaging materials made from mushrooms are insulating, fire-retardant, moisture-resistant, vapour resistant and can absorb mechanical and acoustic impacts. It can be used for shipping food and perishable items.

ii. In cooking process

Polypore mushrooms, such as *Fomes fomentarius*, *Schizophyllum* and *Daldinia* species have been used as fire starters and are referred to as tinder fungi (they catch and hold coals very tightly). Their presence on dead decaying woods also increases burning quality. Tinder is usually prepared from trama, the inner fibrous portion of the mushroom is usually cinnamon in colour. Once trama is obtained, it is ignited and allowed to smoulder for a while. Combustion is remarkably high and materials do not flame and thus cooks food faster. To ignite fire using tinder, agrowastes including dried grass, dried bracken fronds or properly prepared bark fibres are added to the smouldering trama.

iii. Bioremediation

Mushrooms play significant role in bioremediation. They have the ability to absorb heavy metals, including radioactives and breakdown highly intractable and persistent substances, including polyaromatic hydrocarbons (PAHs).

iv. Mycofiltration

The use of mushroom mycelium is now being introduced in water treatment technologies. Their incorporation into engineered water treatment ecosystems and biofiltration media has demonstrated improvements in bacteria reduction and heavy metal removal both in the laboratory and scale demonstrations.

v. Source of pigments

Mushrooms can be used for dyeing wool and other natural fibres. It is used as textile colourant before the advent of synthetic dyes. The chromophores of mushroom are organic and can produce strong and vivid colours of the spectrum. Mushroom pigment may vary with age and undergo distinctive changes on bruising; a characteristic which is used in their identification. Furthermore,

it protects the organism from UV damage and bacterial attack and also acts as insect attractant. Colour from *Pycnoporus cinnabarinus* has been used as hair dye and lipsticks by maidens of Okomu people of Edo State, Nigeria.

vi. Source of flavour

The aroma of some mushroom species play a role in stimulating appetite and gives mushroom dish its characteristic flavour. About 150 flavour compounds have been identified in different mushroom species and can have applications in food, pharmaceutical and cosmetic industries. The amino acids, nucleotides, some other elements such as nitrogen, phosphorus, potassium, sulphur, iron and zinc, and the auto-oxidation of unsaturated fatty acids present in mushrooms have characteristic flavouring role.

vii. Production of enzymes

Mushrooms are able to produce relevant hydrolytic (cellulases and hemicellulases) and oxidative (ligninolytic) extracellular enzymes required to degrade the major components of lignocellulosic biomass into low molecular weight compounds that can be assimilated for fungi nutrition. They produce enzymes such as lignin and manganese peroxidase, laccase and xylanases that are associated with lignin-degrading ability. Enzymes such as xylase from *Pleurotus eryngii* and *Flamulina velutipes*, cellulase from *Stereum ostrea* and *Pleurotus*, glucoamylase and α -amylase from the mycelium of *Tricholoma matsutake*, laccase from *Hypsizygus ulmarius*, fibrinolase from *Schizophyllum commune* and catalase from *Agaricus bisporus* have also been reported.

viii. Bioconversion of agro-industrial wastes

Mushrooms have also played valuable roles in the bioconversion of numerous agro-industrial wastes which pose threat to the lithosphere and hydrosphere, into protein-rich edible mushrooms. The bioconversion of wastes into edible mushrooms provided partial solution to environmental pollution and provided jobs for some youths in Nigeria. Spent mushroom substrates can also be exploited as soil conditioners, single-cell protein, animal feed and biofuel.

ix. Biogeochemical cycling of elements

Mushrooms play a significant role in the biogeochemical cycling of elements as they secrete extracellular enzymes to degrade ligneous substrates on which they grow. Hence, they break down the polycyclic organic polymers in dead woods to release inorganic carbon, nitrogen, sulphur and phosphorus which in turn enhance soil fertility.

Artificial Cultivation of Mushroom

Processes involved in artificial cultivation of mushroom include:

(a) **Tissue culture:** Tissue culture is done from mature mushroom fruitbody gills and propagated on suitable artificial medium usually Potato Dextrose Agar or Sabouraud Dextrose Agar. Subculturing of the mushroom mycelia is done until a pure culture is obtained (Figure 3).

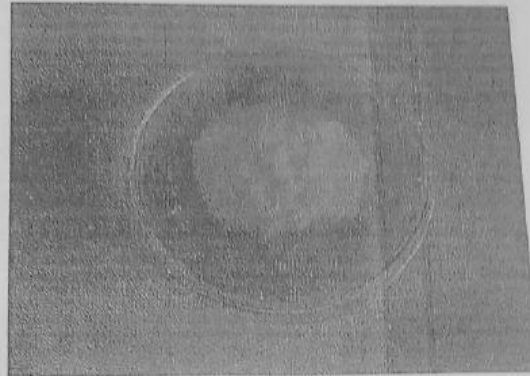


Figure 3: Radial growth of mushroom mycelium on Sabouraud Dextrose Agar.

(b) **Spawn production:**

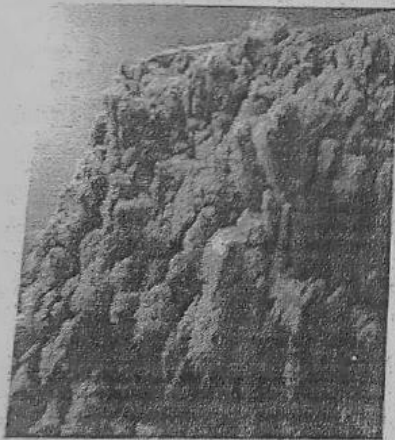
Spawn is a mass of fungal mycelium prepared on a growth medium, usually solid. Different substrates like sorghum, rice straw, millet, sawdust etc. had been used in the production of spawn. Spawn substrate is washed 2 to 3 times and parboiled in tap water. After this, the substrate is rinsed 3 times to remove starch or any other form of debris and allowed to drain in a strainer. The pH of the substrate is adjusted to a range between 6 and 9 by the addition of calcium carbonate depending on the mushroom species. The substrate is two-third filled into autoclavable bottles, plugged with cotton wool and foil paper, sterilized for 15 minutes at 121 °C and left to cool. Pure culture of the mushroom mycelium is transferred from the agar plate and mixed with the substrate in the autoclavable bottle and incubated at 25 °C until the grains are fully ramified with the mushroom mycelium. The fully colonized substrate is called the mother spawn (Figure 4).



Figure 4: Fully colonized spawn bottles of mushroom mycelium.

(c) **Collection and preparation of growth substrate:** Choice of growth substrate used in mushroom cultivation depends on the grower, the availability of the substrate and the potential of a substrate to produce good quantity and quality of mushroom. Various agricultural by-products for example, banana leaves, peanut hull and corn leaves, mango fruits and seeds, sugarcane leaves, wheat, rice straw, chopped cocoa pods, cotton waste, dried chopped maize straw, oil palm (fibre and bunch) wastes, tobacco straw, used tea leaves, rice straw, sugarcane bagasse, newsprint, old rags and sawdust are being used as substrates for mushroom cultivation. Combination of different substrates are also used.

Big sized substrate e.g. corncob or palm bunch are chopped into smaller sizes using cutlass, scissors or mortar or pestle (Figure 5 a&b). Substrate is pasteurized for 30 to 45 minutes and pH is adjusted to 6 to 9 depending on the mushroom species by the addition of calcium carbonate. Using a plastic strainer, the pasteurized substrate is drained and allowed to cool. The substrate is properly mixed and the moisture content is adjusted to about 65%.



a



b

Figure 5: Shredded mushroom substrate (a) Corn cob (b) Paper waste

(d) **Cultivation / Incubation method:** A diversity of cultivation methods utilizing log, shelf, box, bag and bottle has been developed. Selection of the right cultivation methods is based on the mushroom variety, market demands and farmers' preferences.

The substrates are inoculated with spawn grains (Figure 6a), mixed together and filled into containers depending on the cultivation method. Holes are punched in the container to allow for drainage of excess water and then incubated for spawn running under complete darkness in the growth room with 70 to 80% humidity and temperature of 25 to 30 °C. During spawn running, the substrate is being colonized with the mushroom mycelium making the substrate appear white (Figure 6 b&c). The number of days taken for complete spawn running i.e. the time taken for complete ramification of the substrate depends on the quality and the quantity of spawn added to the substrate.



Figure 6: The incubation method in artificial cultivation of mushroom (a) mixture of wood shavings and spawn grain in bowl (b) waste paper fully colonized with *Pleurotus florida* mushroom mycelium after spawn running (c) corn cob fully colonized with *Pleurotus florida* after spawn running

(e) **Fruiting and harvesting:** After incubation and spawn running, substrates are removed from the incubation room to the fruiting room which has good lighting and ventilation. In cases where natural ventilation does not suffice, air can be supplied using exhaust fan. Humidity of bags can be maintained by watering three times daily. Pin holes are made in the bags to exhaust gases. Fruiting starts with the formation of pinhead which is the immature mushroom fruitbody (Figure 7). As soon as fruitbodies developed and attain maturity (Figure 8), they are cut just above the

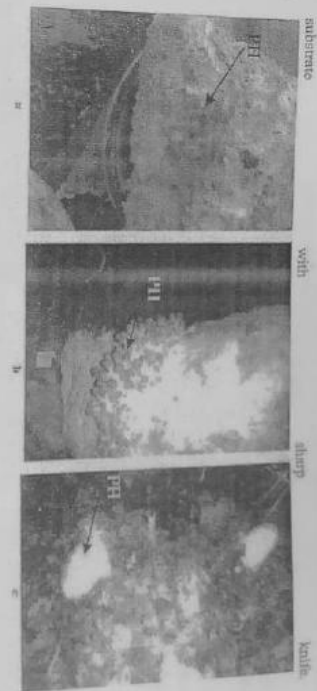


Figure 7: Pinhead formation on different substrates. PH – Pinhead (a) pinhead formation of *Pleurotus florida* on corn cob substrate (b) pinhead formation of *Pleurotus florida* on old newspaper substrate (c) pinhead formation of *Pleurotus florida* on wood shred substrate.

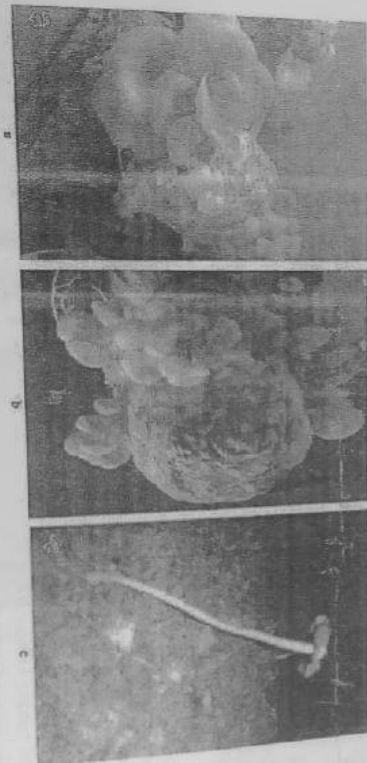


Figure 8: *Pleurotus florida* mushroom fruitbodies on different substrates. (a) fruitbodies of *Pleurotus florida* mushroom on corn cob. (b) fruitbodies of *Pleurotus florida* mushroom on old newspaper substrate. (c) low quality fruitbody of *Pleurotus florida* mushroom on wood shred.